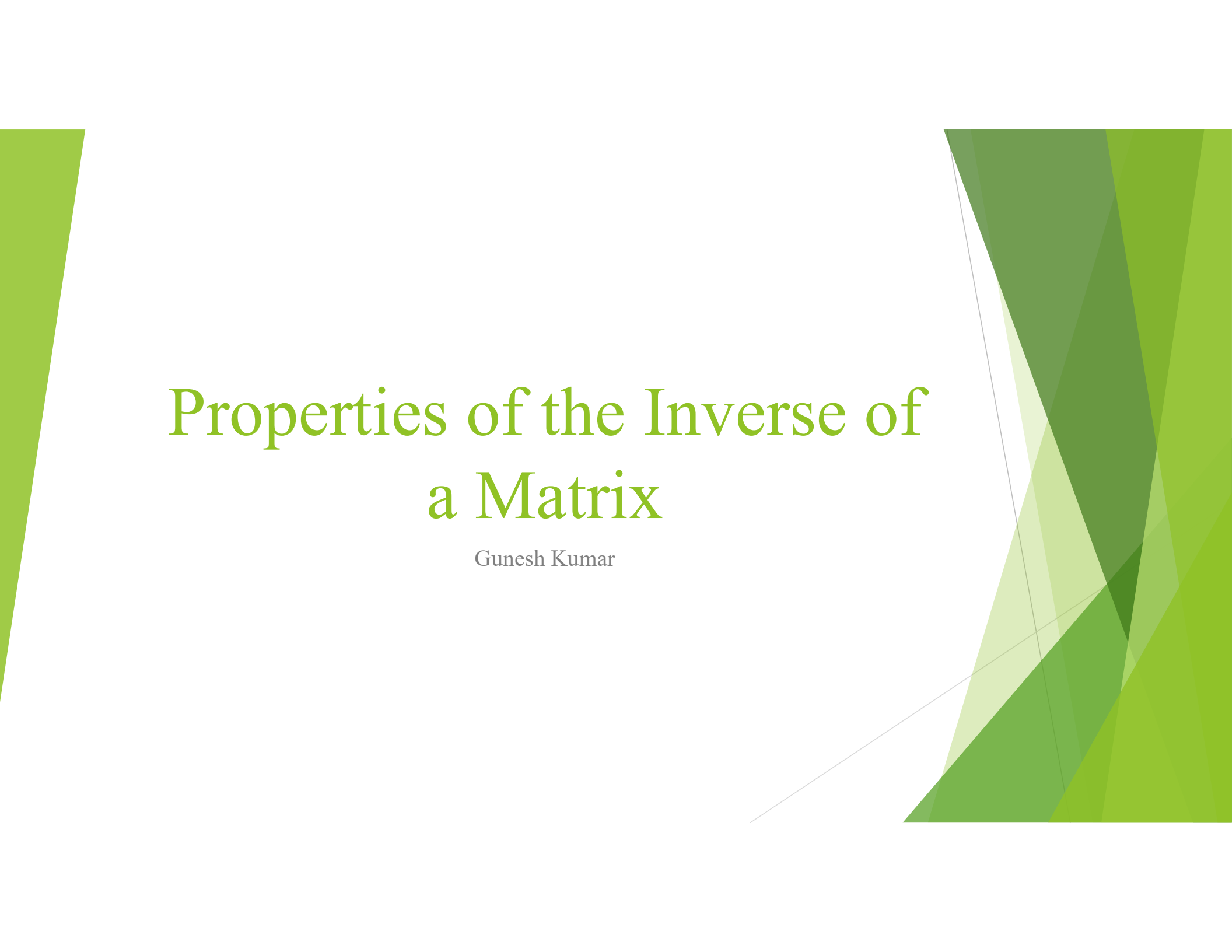




Properties of the Inverse of a Matrix

Gunesh Kumar

Properties of Inverse of a Matrix

It is easy to verify the following properties of inverses of matrices

- (i) For any invertible matrix A , $(A^{-1})^{-1} = A$
- (ii) For nonsingular matrices A and B , $(AB)^{-1} = B^{-1}A^{-1}$
- (iii) For any invertible matrix A , $(A^T)^{-1} = (A^{-1})^T$

•

Q#1: If $A^{-1} = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 4 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0 & -2 \\ 1 & 1 & 1 \end{bmatrix}$

Compute $(AB)^{-1}$.

Sol:

We Know that

$$(AB)^{-1} = B^{-1}A^{-1}$$

- $$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0 & -2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 1 + 1 \times 0 + 1 \times 1 & 2 \times 3 + 1 \times 1 + 1 \times (-1) & 2 \times 0 + 1 \times 1 + 1 \times 4 \\ 0 \times 1 + 0 \times 0 + (-2) \times 1 & 0 \times 3 + 0 \times 1 + (-2) \times (-1) & 0 \times 0 + 0 \times 1 + (-2) \times 4 \\ 1 \times 1 + 1 \times 0 + 1 \times 1 & 1 \times 3 + 1 \times 1 + 1 \times (-1) & 1 \times 0 + 1 \times 1 + 1 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 + 0 + 1 & 6 + 1 - 1 & 0 + 1 + 4 \\ 0 + 0 - 2 & 0 + 0 + 2 & 0 + 0 - 8 \\ 1 + 0 + 1 & 3 + 1 - 1 & 0 + 1 + 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 5 \\ -2 & 2 & -8 \\ 2 & 3 & 5 \end{bmatrix}$$

•
Q#2: Use the inverse coefficient matrix, solve the system of linear equations

$$3x_1 + 2x_2 + 4x_3 = 7$$

$$2x_1 + x_2 + x_3 = 4$$

$$x_1 + 3x_2 + 5x_3 = 3$$

Sol:

We have non-homogeneous system of linear equations $Ax = b$, then the solution is $x = A^{-1}b$. So first we find the A^{-1}

Here, $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix}$

To find A^{-1} we start with the augmented matrix $[A|I]$.

$$[A|I] = \left[\begin{array}{ccc|ccc} 3 & 2 & 4 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 \\ 1 & 3 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 & 1 & 0 \\ 3 & 2 & 4 & 1 & 0 & 0 \end{array} \right] \text{ by } R_{13}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 0 & -5 & -9 & 0 & 1 & -2 \\ 0 & -7 & -11 & 1 & 0 & -3 \end{array} \right] \begin{array}{l} \text{by } R_2 - 2R_1, \\ R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 0 & 2 & 2 & -1 & 1 & 1 \\ 0 & -7 & -11 & 1 & 0 & -3 \end{array} \right] \text{ by } R_2 - R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 0 & 1 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -7 & -11 & 1 & 0 & -3 \end{array} \right] \text{ by } \frac{1}{2}R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 0 & 1 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -4 & -\frac{5}{2} & \frac{7}{2} & \frac{1}{2} \end{array} \right] \text{ by } R_3 + 7R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 0 & 1 \\ 0 & 1 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{5}{8} & -\frac{7}{8} & -\frac{1}{8} \end{array} \right] \text{ by } \left(-\frac{1}{4}\right)R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & -\frac{25}{8} & \frac{35}{8} & \frac{13}{8} \\ 0 & 1 & 0 & -\frac{9}{8} & \frac{11}{8} & \frac{5}{8} \\ 0 & 0 & 1 & \frac{5}{8} & -\frac{7}{8} & -\frac{1}{8} \end{array} \right] \text{ by } \begin{array}{l} R_1 - 5R_3 \\ R_2 - R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{8} & \frac{2}{8} & -\frac{2}{8} \\ 0 & 1 & 0 & -\frac{9}{8} & \frac{11}{8} & \frac{9}{8} \\ 0 & 0 & 1 & \frac{5}{8} & -\frac{7}{8} & -\frac{1}{8} \end{array} \right] \text{ by } R_1 - 3R_2$$

$$A^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ -\frac{9}{8} & \frac{11}{8} & \frac{5}{8} \\ \frac{5}{8} & -\frac{7}{8} & -\frac{1}{8} \end{bmatrix}$$

•
Now

$$x = A^{-1}b$$

$$x = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ -\frac{9}{8} & \frac{11}{8} & \frac{5}{8} \\ \frac{5}{8} & -\frac{7}{8} & -\frac{1}{8} \end{bmatrix} \begin{bmatrix} 7 \\ 4 \\ 3 \end{bmatrix}$$

$$x = \begin{bmatrix} \frac{7}{4} + \frac{4}{4} - \frac{3}{4} \\ -\frac{63}{8} + \frac{44}{8} + \frac{15}{8} \\ \frac{35}{8} - \frac{28}{8} - \frac{3}{8} \end{bmatrix} = \begin{bmatrix} \frac{8}{4} \\ -\frac{4}{8} \\ \frac{4}{8} \end{bmatrix} = \begin{bmatrix} 2 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

•
Q#3: Use Gauss Elimination Method to solve the system of linear equations

$$x + y + z = 3$$

$$2x + 2y + 2z = 6$$

$$x - y + z = 1$$

Solution:

Step#1:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}$$

Step#2:

$$A_b = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & 1 & 1 \end{array} \right]$$

•
Step#3:

$$A_b \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 \end{array} \right] \text{ by } \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right] \text{ by } \left(-\frac{1}{2} \right) R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ by } R_{23}$$

•
Step#4:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$
$$x + y + z = 3 \rightarrow (i)$$
$$y = 1 \rightarrow (ii)$$

Put $y = 1$ in eq: (i)

$$x + 1 + z = 3$$
$$x = 2 - z$$

Let $z = a, a \in R$, then

$$x = 2 - a$$
$$y = 1$$
$$z = a$$

This system of equation has infinite solution

•
Q#4: Use Gauss Elimination Method to solve the system of linear equations

$$\begin{aligned}x + 2y - z &= 4 \\ 2x + 4y - 2z &= 9 \\ 3x - y + z &= 2\end{aligned}$$

Solution:

Step#1:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix}$$

Step#2:

$$A_b = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 4 & -2 & 9 \\ 3 & -1 & 1 & 2 \end{array} \right]$$

- Step#3:

$$A_b \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & -7 & 4 & -10 \end{array} \right] \text{ by } \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -\frac{4}{7} & \frac{10}{7} \end{array} \right] \text{ by } \left(-\frac{1}{7} \right) R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & -\frac{4}{7} & \frac{10}{7} \\ 0 & 0 & 0 & 1 \end{array} \right] \text{ by } R_{23}$$

•
Step#4:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -\frac{4}{7} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{10}{7} \\ 1 \end{bmatrix}$$
$$x + 2y - z = 4$$
$$y - \frac{4}{7}z = \frac{10}{7}$$
$$0 \neq 1$$

Here is contradiction so this system of equations has no solution.